A hybrid framework for short-term wind speed forecasting based on EEMD and LSTM Network

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Abstract - Renewable energy resources provide reliable power supply and fuel diversification. This increases energy security and decreases the need for fossil fuels. Renewable energy resources are eco- friendly and free from polluting nature for generating electricity. In recent years, wind energy is being considered as one of the important sustainable energy resources. As it is eco-friendly, carbon-free, and non- emission of harmful gases. Wind energy, which is intermittent by nature has a remarkable power system operation, grid security. Accuracy and reliable wind speed forecasting are crucial for the operation of power grids. To improve wind speed prediction accuracy, a huge number of forecasting approaches have been proposed. In this project, a combined model is implemented using Ensemble Empirical Mode Decomposition (EEMD) and long short-term memory (LSTM) network for accurate wind speed prediction. In this model, the EEMD method is used to decompose the raw data (collected from wind farm) of wind speed time series into several IMF's (Intrinsic Mode Function). Then the LSTM network is employed to predict the wind speed. This proposed hybrid deep learning model is completely validated with real-time data. By using statistical error values, the performance is analyzed. Finally, the proposed model gives high-ranking performance when compared with conventional approaches.

Keywords -EEMD (Ensemble Empirical Mode Decomposition), LSTM (Long short-term Memory), Machine Learning, Hybridframework, Prediction.

I. INTRODUCTION

Renewable energy plays a crucial role in the global energy contribution, and it is an appropriate and viable solution for supplying power to the un-electrified areas and to feed the existing power system. This wind energy system, which is irregular or not continuous by nature, has a remarkable impact on the operation of the power system, market economics, and electric grid security. Accurate wind energy forecasting is crucial in the optimal scheduling of wind farms. By using the wind, the energy of wind power is obtained. By increasing the accuracy of forecasting the operators of the power system increases the continuity of power supply to the consumers. Sustainable energy resources have a crucial role to meet future power energy needs in both urban and rural areas. A high priority should be given to renewable energy resources for the development and utilization of electricity by considering the adverse environmental impacts of fossil-fuels generation. The requirement for renewable energy resources is increasing rapidly globally. Human activities are directly related to increasing atmospheric levels of global warming gases such as (CO_2) carbon dioxide, Chloro Fluoro Carbons (CFCs), methane, and carbon monoxide (CO). There also exists a close relation between global warming that involves greenhouse gasand some environmental problems.

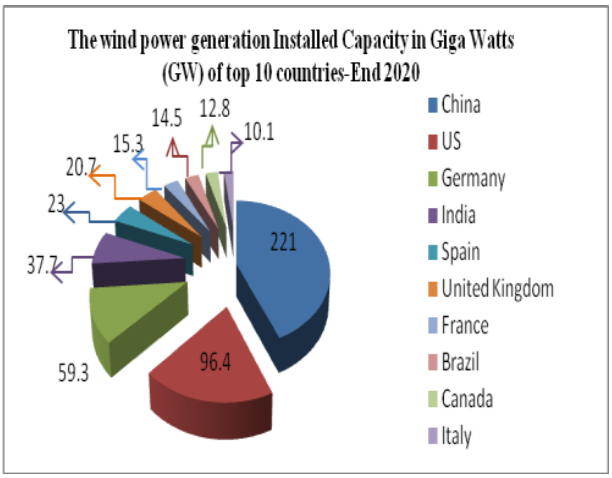


Fig. 1. The wind power generation installed capacity in gigawatts (GW)of top 10 countries at the end of the year 2020.

Forecasting accuracy is a critical component in the quest for further growth and development of low-carbon economies and the conversion to sustainable energy sources on a global scale. Within this landscape, the wind power industry emerges as a pivotal player. Its significance lies not only in its contribution to renewable energy generation but also in its role in bolstering the stability and efficiency of energy systems. Precise wind forecasting stands as a linchpin in ensuring the secure operation of power grids and facilitating the seamless integration of wind power into existing energy infrastructures.

To appreciate the magnitude of the wind power sector's impact, it's imperative to examine the installed wind energy capacity across key nations. As of the year-end of 2020, China leads the pack with an impressive 221 gigawatts (GW) of renewable energy generation, firmly establishing itself as a global leader in sustainable energy initiatives. Following closely behind is the United States, boasting a substantial 96.4 GW, indicative of its significant strides in renewable energy adoption. Germany secures the third spot with 59.3 GW, reflecting its commitment to transitioning towards cleaner energy sources.

India, Spain, the United Kingdom, France, Brazil, Canada, and Italy round out the top 10, each contributing significantly to the global wind energy landscape. From India's 37.7 GW to Italy's 10.1 GW, these nations collectively underscore the widespread adoption of wind power as a viable alternative to traditional fossil fuels.

Moreover, the trajectory of wind energy installation over recent years further highlights the sector's momentum. From 2019 to 2023, there has been a consistent year-on-year increase in wind installation, a trend that signifies both the growing demand for renewable energy and the industry's capacity for expansion and innovation.

In essence, the prominence of wind power in the global energy transition is undeniable. As nations strive to meet ambitious carbon reduction targets and embrace sustainable practices, the wind power industry stands ready to lead the charge towards a cleaner, greener future. Through continued investment in technology, research, and collaborative efforts, the potential for wind power to drive meaningful change on a global scale remains boundless.

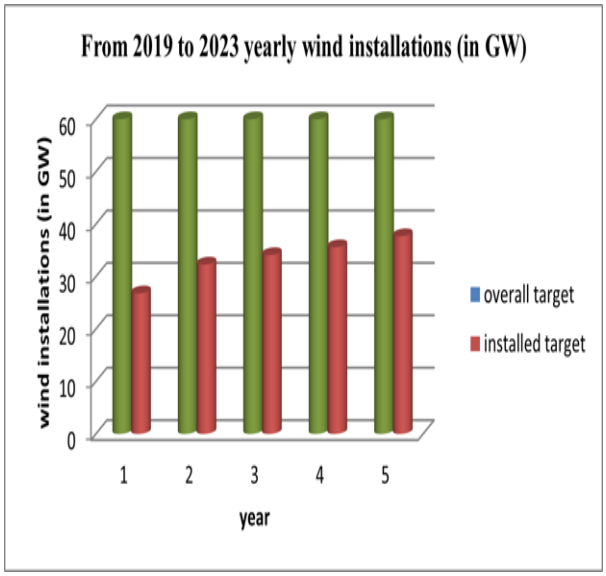


Fig. 2. From 2019 to 2023 yearly wind installations (in GW)

II. DEVELOPED HYBRID MODEL

This section presents the EEMD technique, LSTM network, and the proposed hybrid EEMD-LSTM approach in detail.

Advantages Of Wind Energy:

- Wind energy eliminates the need for fuel provision and transportation, reducing dependency on fossil fuels and associated emissions.
- Wind energy systems often have moderate to low startup costs compared to traditional energy sources, making them accessible to a wider range of users.
- The land beneath wind turbines can be utilized for other activities, maximizing land use efficiency, and potentially offering additional economic benefits.
- Wind turbines efficiently convert kinetic energy from the wind into electricity, providing a reliable source of renewable energy.
- Wind energy production does not emit CO2 or other greenhouse gases, contributing to a cleaner and healthier environment.
- The wind is a secure energy resource available in many locations worldwide, offering energy independence and resilience against supply disruptions.

A. EEMD Decomposition technique

The Empirical Mode Decomposition (EMD) technique, coupled with Ensemble Empirical Mode Decomposition (EEMD), stands as a powerful tool in signal processing and time series analysis. EEMD, an extension of EMD, addresses some of the limitations of traditional EMD, particularly in handling non-stationary and noisy data.

At its core, EMD is a data-driven method that decomposes a given signal into a finite set of oscillatory components known as Intrinsic Mode Functions (IMFs). These IMFs capture the inherent oscillatory modes present within the signal, providing a localized and adaptive representation of the data. However, EMD's effectiveness can be hindered by issues such as mode mixing and boundary effects, especially in the presence of noise.

EEMD overcomes these challenges by introducing an ensemble approach. Instead of relying on a single realization of noise-assisted data, EEMD generates multiple noise realizations and applies EMD to each, resulting in a collection of IMFs.

These IMFs are then averaged across the ensemble to obtain the final decomposition. By averaging out the noise across multiple realizations, EEMD enhances the robustness and reliability of the decomposition process, making it well-suited for analyzing complex and noisy signals.

One of the key advantages of EEMD is its ability to adaptively decompose non-stationary signals with varying frequencies and amplitudes. This adaptability makes it particularly useful in analyzing time series data, where traditional frequency-domain methods may fall short.

Additionally, EEMD's data-driven nature eliminates the need for predefined basis functions or model assumptions, allowing for a more flexible and versatile approach to signal decomposition.

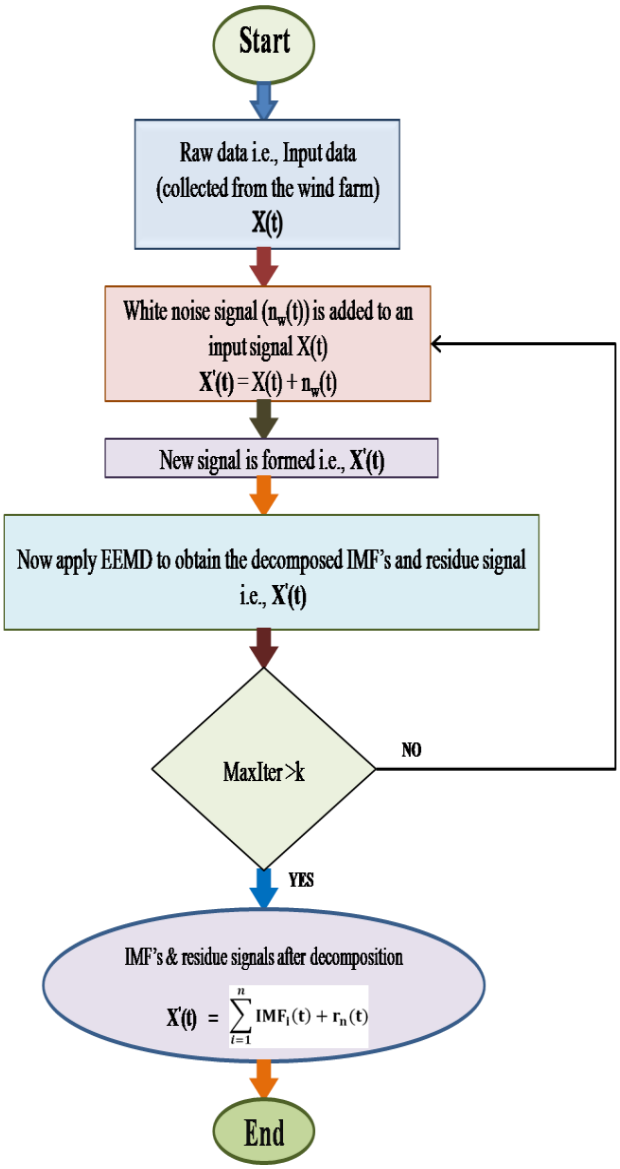


Fig. 3. Flow-chart of EEMD

EMD is called Empirical mode decomposition, EMD algorithm technique (adaptive method) is applied for analyzing non-linear and non-stationary data into number of intrinsic modes of oscillation.

The mode mixing problem is encountered by EMD algorithm regularly, by the process of decomposing the non- linear and non-stationary signal into several IMF components. So, to avoid this problem EEMD (Ensemble Empirical mode decomposition) is proposed.

EEMD is called (Ensemble Empirical mode decomposition). It is a noise-assisted data analyst method that defines the actual IMF components as the mean of an N ensemble of each trial by adding a white noise signal (nw(t)) of finite-magnitude to decompose the results of the signal. The white noise signal (nw(t)) of finite-magnitude is added to the decomposition which results in a new signal that divides into different signals that are IMF's and residue signals.

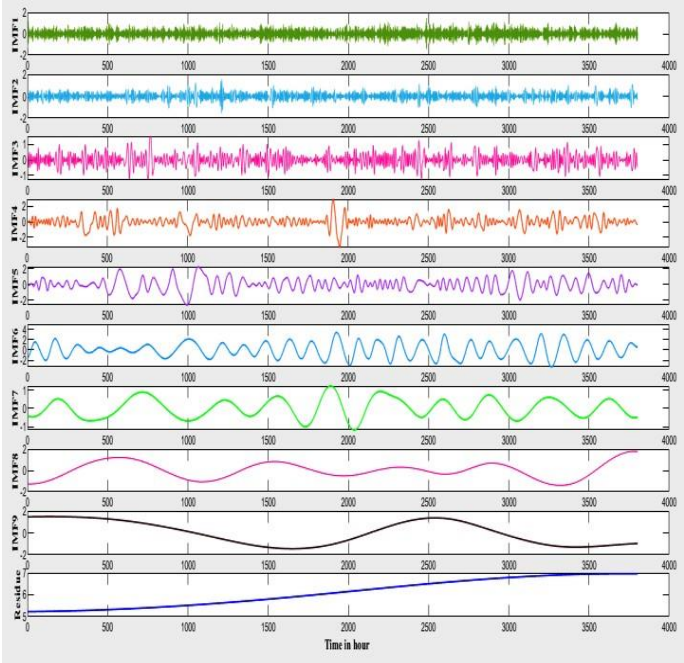


Fig. 4. Number of IMF's and residue signals obtained by the EEMD algorithm.

The raw data i.e., wind speed data collected from the wind powerplant. For demonstrating the performance of the forecasting methods, the raw data collection was one-hour data samples. The raw data containing 3750 samples are given to the EEMD algorithm for extracting the Intrinsic Mode Function's (IMF's) and a residual signal $r_n(t)$. The above Fig. 4. shows the number of IMF's and residue signals which were obtained by the EEMD (Ensemble Empirical Mode Decomposition) procedure.

B. LSTM (Long short-term Memory) Network

In LSTM network, memory cell is integrated into hidden layers neurons. LSTM is a chain-like structure that permit it to contain information for a long period of time. LSTM is a kind of recurrent neural network (Hidden layer neurons have a self-connection and have memory, at any point hidden layer neuron collects activation from the lower layers as well as previous activation value and output is fed back to inputuntil the output is of the desired output) proposed to avoid vanishing gradient difficulty and can keep the data for additional extended period.

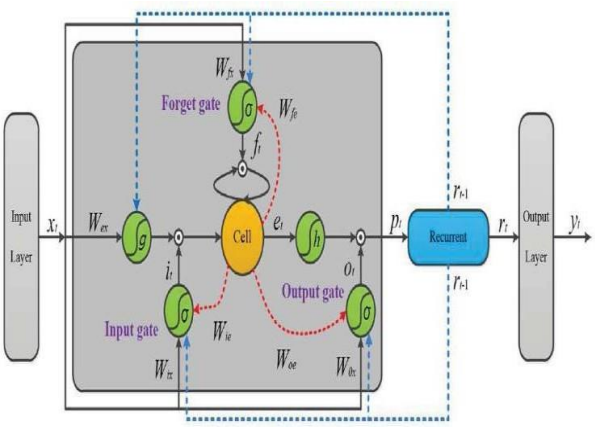


Fig. 5. LSTM Neural Network

Input gate: $i_t = \sigma(W_{ix}x_t + W_{ip}P_{t-1} + W_{ie}e_{t-1} + b_i)$,
Output gate: $o_t = \sigma(W_{ox}x_t + W_{op}P_{t-1} + W_{oe}e_{t-1} + b_o)$,
Forget gate: $f_t = \sigma(W_{fx}x_t + W_{fp}P_{t-1} + W_{fe}e_{t-1} + b_f)$,
Temporary cell state: $\bar{e}_t = g(W_{ex}x_t + W_{ep}P_{t-1} + b_e)$,
Cell state: $e_t = i_t \odot \bar{e}_t + f_t \odot e_{t-1}$,
Output: $P_t = o_t \odot h(e_t)$
Output layer: $y_t = \phi(W_{yt}P_t + b_y)$,
Input vector is x_t,
Output vector is y_t,
Input gate is i_t, Output gate is o_t, forget gate is f_t, Temporary state is $\bar{e}_t$, finishing state is e_t,
Output of memory block is P_t,
σ indicates gate activation function, g indicates input activation function,
h indicates output activation function, $\odot$ is element-wise multiplication between two vectors,
ϕ is output activation function of LSTMs,
W_{ix}, W_{ip}, W_{ie}, W_{ox}, W_{op}, W_{oe}, W_{fx}, W_{fp}, W_{fe}, W_{ex}, W_{ep}, W_{yp} indicates weight matrices,
b_i, b_o, b_f, b_e, and b_y are bias vectors.

The sigmoid function (σ) is used for gate activation function and in most of the cases, hyperbolic function tangent function is used for input and output activation functions. The flow-chart of hybrid framework EEMD-LSTM network machine learning model is shown in above Fig. 6.

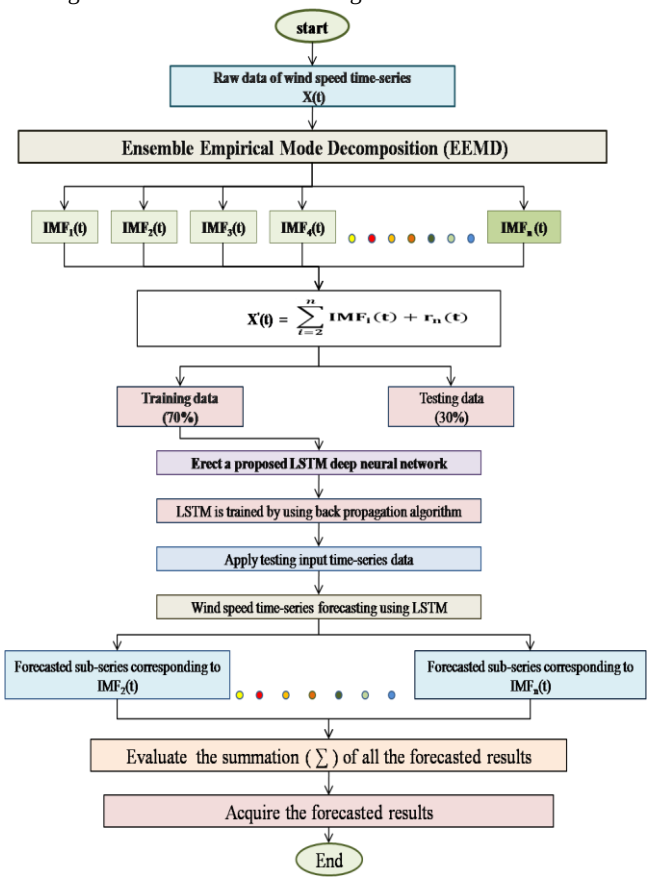


Fig. 6. Flow-chart of hybrid framework EEMD-LSTM Network

A. Developed Hybrid Framework model.

LSTM Network is trained by using the input data, which is obtained by EEMD decomposed wind speed data of IMF's and residual signals. For accurate wind forecasting IMF1(t) is eliminated before applying to the LSTM network and considered the remaining IMF's and residual signals ($r_n(t)$). To obtain the predicted outputs the testing data (30%) is applied to the trained LSTM network approach. The total dataset(input data) contains 3750 data samples, out of them 2626 data samples were used for training(70%) and 1124 data samples were used for testing(30%). The results were approached by the developed hybrid framework i.e., EEMD and LSTM network by the total sum of the forecasted outputs. The general flow-chart of the developed approach i.e., the hybrid framework of EEMD and LSTM Network is shown in Fig. 7.

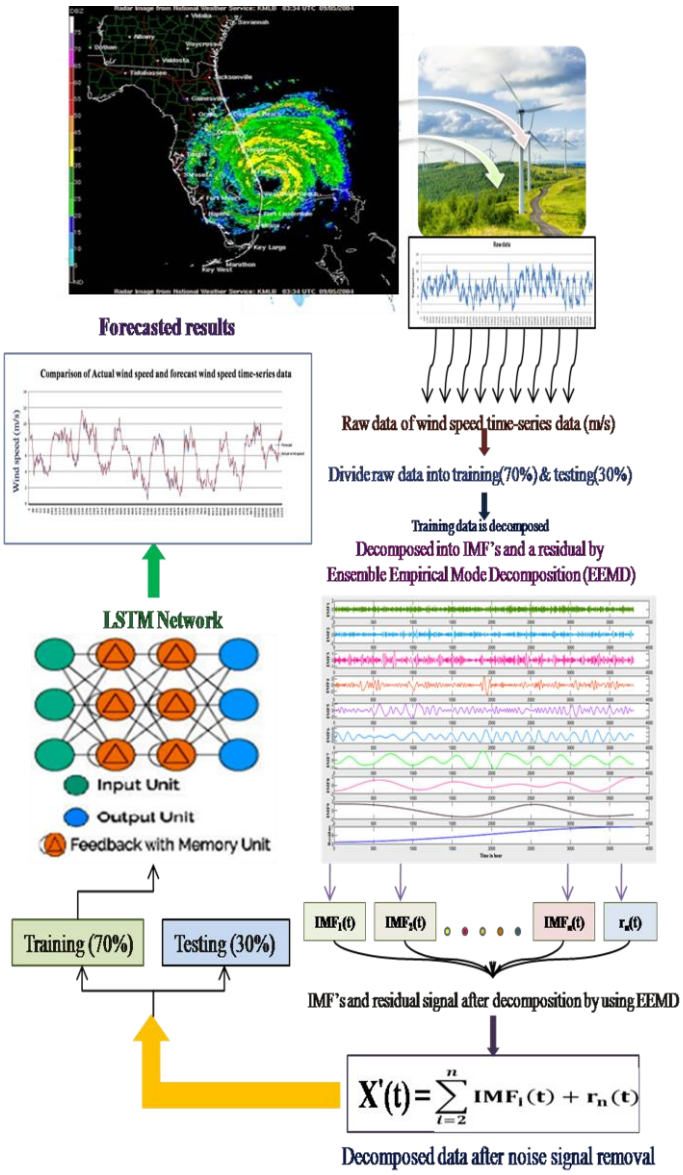


Fig. 7.General flow-chart of developed model hybrid framework EEMD and LSTM network

III. RESULTS AND DEVELOPED MODEL VALIDATION

The four evaluation criteria are considered to evaluate forecast capacity of different approaches and developed model to analyze performance of the existing and proposed approach. Smaller values in the evaluation criteria reflect better forecasting performance.

TABLE I

Sl. no	Metric	Abbreviation	Equation
1.	MAE	Mean absolute error	$\frac{1}{N}\sum_{i=1}^{N}\lvert(Ypred - Ytest)\rvert$
2.	SSE	Square Sum of the Error	$\sum_{i=1}^{N}(Ypred - Ytest)^2$
3.	RMSE	Root-mean-square error	$\sqrt{\sum_{i=1}^{N}(Ypred - Ytest)^2}$
4.	MAPE(%)	Mean absolute percentage error	$MAPE = \frac{1}{N}\sum_{i=1}^{N}\frac{\lvert Ypred-Ytest\rvert}{\lvert Ytest\rvert}*100\%$

Table I:-The statistical error measures used for error evaluation Where, Ypred is the forecasted values. Ytest is the actual wind speed values. N is the sample size.

TABLE II

MODELS	MAE	SSE	RMSE	MAPE(%)
FFBP	0.6146	739.2622	0.8106	13.3811
ELMAN	0.6009	727.7886	0.8043	13.0106
LSTM	0.4719	441.926	0.62703	10.4151
EEMD-FFBP	0.2735	126.4154	0.3352	5.7081
EEMD-ELMAN	0.2748	127.4983	0.3366	5.7389
EEMD-LSTM	0.1354	35.8471	0.1786	2.7723

Table II:- Statistical error values of all existing and developed approaches

The above Table II shows the statistical error values of all existing and developed i.e., the hybrid framework of EEMD and LSTM network approaches. When compared to existing approaches i.e., FFBP, ELMAN, LSTM, EEMD-FFBP, EEMD-ELMAN, the proposed approach i.e., the hybrid framework of EEMD-LSTM gives the high-ranking performance. The table.6.4 shows the fewer error value of the developed hybrid framework of the EEMD-LSTM network approach when compared to the remaining approaches. The MAE error value of LSTM is 0.4719, which is the superior value when compared with the 0.6146, 0.6009 of FFBP and ELMAN, respectively. Similarly, the RMSE error value of LSTM is 0.62703, which is the superior error value when compared with the 0.8106, 0.8043 of FFBP and ELMAN, respectively. Similarly, the SSE and MAPE index value of LSTM is a better value in comparison with the FFBP and ELMAN, respectively.

The pre-processing algorithm which is noise-assisted data, such as the EEMD algorithm can upgrade the forecasted performance by eliminating the noise signal from the wind. speed data of benchmark approaches. Among all existing approaches, EEMD-LSTM gives a high-ranking performance, superior, and obtained less error value. The superior values of MAE, SSE, RMSE, and MAPE(%) value obtained using the developed EEMD-LSTM network approach are 0.1354, 35.8471, 0.1786, and 2.7723(%), respectively.

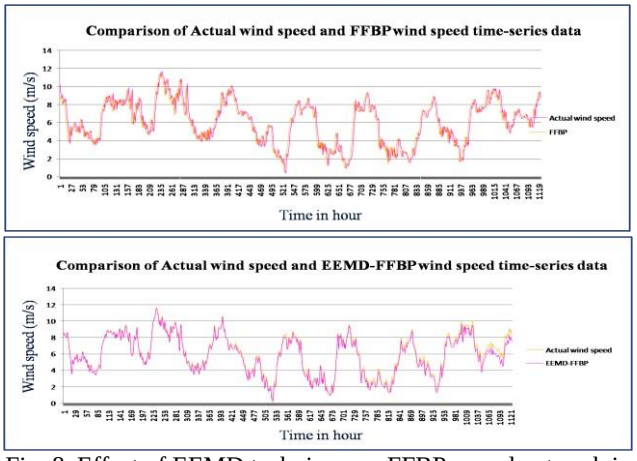


Fig. 8. Effect of EEMD technique on FFBP neural network in forecasting results

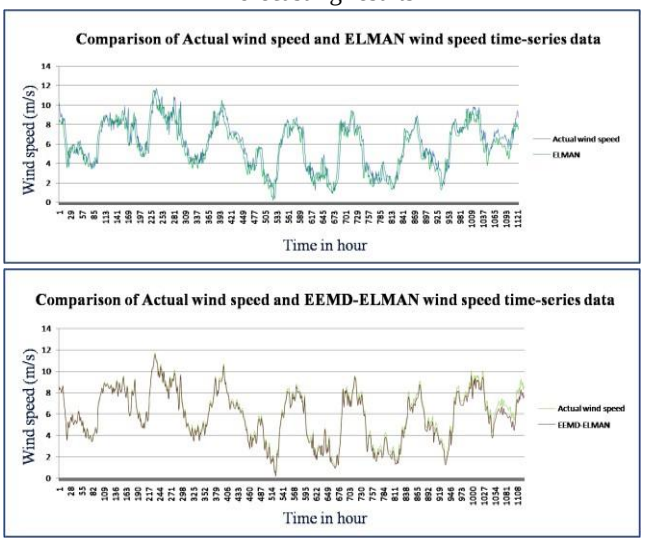


Fig. 9. Effect of EEMD technique on ELMAN neural network in forecasting results

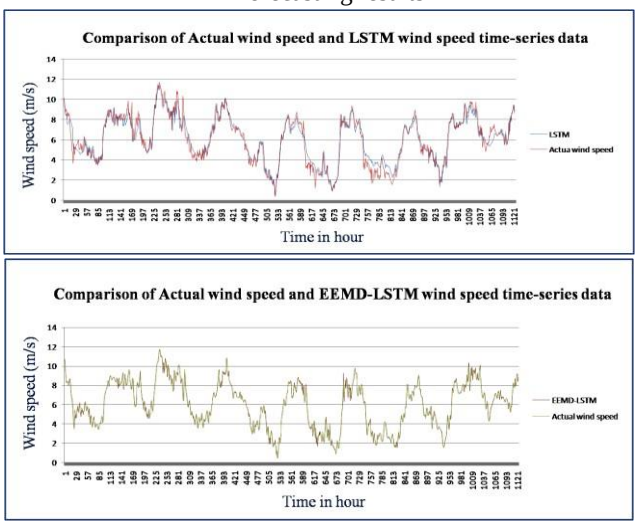
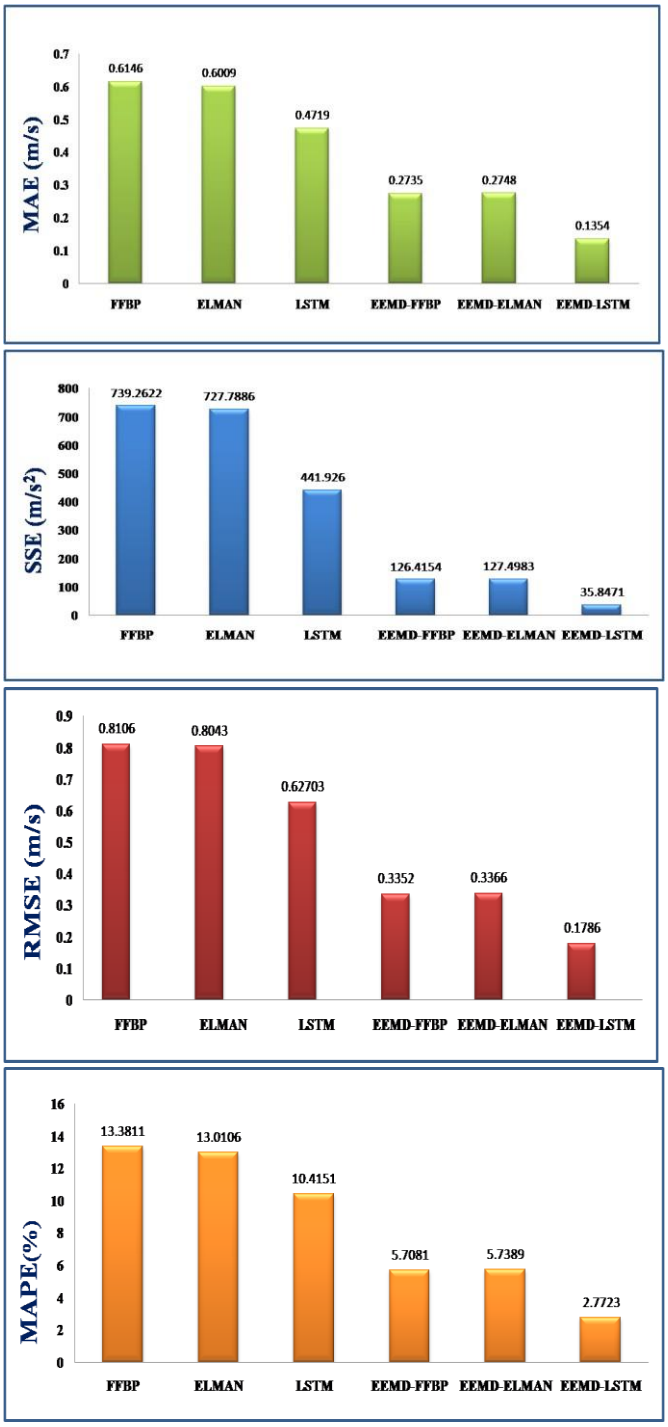


Fig. 10. Effect of EEMD technique on LSTM neural network in forecasting results

By comparing the below fig's i.e., 8, 9, 10 effect of EEMD technique among all existing approach and developed approach in forecasting results, the proposed approach the hybrid framework of EEMD-LSTM network gives accuracy, reliable, superior forecasted outputs.

Below Fig. 11. Shows the comparison of statistical error measures of all benchmark & proposed approaches. Therefore, by comparing the statistical error measures the developed approach gives less error value and higher-ranking performance among all the benchmark approaches.

Fig. 11. Comparison of statistical error measures of existing and proposed approaches.



IV. CONCLUSION

As the years pass, wind speed forecasting has increased massively and played a significant key in the wide-ranging integration of wind power generation. As wind energy is intermittent which makes the prediction challenge. To enhance the accuracy and reliability of short-term wind forecasting the approach EEMD-LSTM is introduced. The hybrid framework EEMD-LSTM network approach is proposed to decrease the forecasting complexity/difficulties of wind speed data. A real-time data sample is taken and conducted an experiment to assess the proposed method on the raw data, which is been collected from the wind farm, NIWE, Telangana, India. The outcomes specify the developed approach gives an accurate, superior, and reliable performance in short-term wind speed prediction the raw data when contrast to all benchmark approaches.

The hybrid framework of EEMD-LSTM aims to address the complexity and difficulties inherent in wind speed data forecasting. By combining the Empirical Mode Decomposition (EEMD) technique with the Long Short-Term Memory (LSTM) network, this approach seeks to improve the effectiveness of wind speed predictions.

To evaluate the performance of the proposed method, a real-time data sample was collected from a wind farm operated by the National Institute of Wind Energy (NIWE). This raw data was then subjected to an experiment to assess the efficacy of the EEMD-LSTM approach.

The outcomes of the experiment indicate that the developed approach offers accurate, superior, and reliable performance in short-term wind speed prediction compared to benchmark approaches. By leveraging the capabilities of EEMD and LSTM, the hybrid framework demonstrates its effectiveness in overcoming the challenges associated with wind speed forecasting.

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